

Shehla Amir

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Research Interests: Wireless communication theory, Fiber Wireless communication

Education

North Carolina State University

PhD in Electrical Engineering

Raleigh, North Carolina

January 2022 - Dec 2026

Advisor: Prof. Robert W. Heath Jr. **CGPA:** 3.39/4.0

Relevant Courses: *Space-time communication theory, Signal processing and deep learning for advanced MIMO systems, LTE and 5G communications, Wireless Communications, Internet Protocols, Electromagnetic Fields, RF Design for wireless, Detection and Estimation theory, Photonics, Optical communications, Information theory*

Lahore University of Management Sciences

Bachelor of Science in Electrical Engineering

Lahore, Pakistan

August 2017 - July 2021

CGPA: 3.69/4.0 **EE GPA:** 3.74/4.0

Awards: Dean's Honor List 2020, 2019, 2018

Relevant Courses: *Electromagnetic Field and Waves, Communication systems, Computer Networks, Deep Learning, Digital Signal Processing, Principles of Optics, Stochastic systems, Advance Digital Signal Processing, Power Electronics*

Skills

Research: Wireless communication, Radio-over-fiber, MIMO Signal Processing, Active/Passive Sensing

High-Level Languages: C, C++, Python

Algorithm Development: MATLAB

Design and Simulation Tools: LTSpice, Proteus, Cadence AWR, HFSS

Technical Tools: mickroC, MPLAB, Adobe Photoshop, Wireshark

Experience

Graduate Research Assistant

North Carolina State University

Jan 2022 - August 2024

Supervisor: Prof. Robert W. Heath Jr.

- Working on combining optical communication theory with wireless communication for future radio access networks
- Utilizing signal processing techniques for the design of efficient optical systems
- Previously worked on joint communication and radar for multi-node radar systems

Research & Development Engineering Intern

Qorvo

May 2023 - August 2023

Supervisor: Dr. Mark Gallagher

- Understanding the working principles of surface acoustic waves for RF circuit design
- Automated the COM model vs. Measurement analysis for surface acoustic waves
- Streamlined the surface acoustic waves-based filter design flow using cadence AWR and HFSS

Research Assistant

Lahore University of Management Sciences

Energy Storage Systems

June 2021 - November 2021

Supervisor: Dr Ijaz Haider Naqvi

- Data-driven solution to estimate the remaining useful life of the battery
- Using the equivalent circuit model of lithium-ion battery to estimate the State of Health of the battery

Teaching Experience

Teaching Assistant

North Carolina State University

Responsibilities: Supported instructor in grading exams and labs. Assisted students in performing the labs and held weekly office hours for any detailed help required.

1. **Analytical Foundations of Electrical and Computer Engineering** taught by Prof. Brian Hughes (Fall 2024)

Teaching Assistant

Lahore University of Management Sciences

Responsibilities: Supported instructor in grading assignments, quizzes, labs and vivas. Assisted students in performing the lab experiments and held weekly office hours and tutorials for any detailed help required.

1. **Communication Systems** taught by Dr. Ijaz Haider Naqvi (Fall 2020-21)
2. **Electromagnetic Fields and Waves** taught by Dr. Muhammad Imran Cheema (Fall 2020-21)
3. **Digital Logic Circuits** taught by Dr. Mohammad Jahangir Ikram (Spring 2020, Spring 2021)

Research Projects

MIMO aided by radio-over-multi-mode-fiber

July 2024 - Ongoing

Supervisor: Prof. Robert Heath Jr.

Motivation: The advancement of RAN-based architectures for future communication networks will require increased fronthaul capacity. While single-mode fiber is the current standard, its capacity is limited. Multi-mode fiber offers greater flexibility and scalability, allowing for higher capacity increases needed to support next-generation technologies.

Actionable Items:

- Linear model analysis for convergence of wireless and optical MIMO channels
- Investigating the increase in capacity networks utilizing various multiplexing techniques in fiber along with wireless channels.

Analog radio-over-fiber for beyond CPRI applications

September 2023 - August 2024

Supervisors: Prof. Robert Heath Jr.

Motivation: Radio-over-fiber (RoF) enables high data rate transmission and network centralization by providing a reliable link between the remote radio head (RRH) and baseband unit (BBU). Current RoF systems use digital components like ADCs and filters for data transmission. Analog RoF (A-RoF) reduces the need for ADCs at the RRH by transmitting analog radio signals directly between the RRH and BBU.

Actionable Items:

- Formulated linear input/output model combining optical and wireless channel for uplink MIMO aided by analog radio-over-fiber
- Investigated the spectral efficiency and power consumption of the MIMO aided by A-RoF and compared to digital architectures
- "Spectral efficiency analysis of uplink MIMO aided by analog radio-over-fiber" for submission to OFC 2025
- "Power consumption and spectral efficiency analysis for uplink MIMO antenna aided by analog radio-over-fiber" draft in progress for submission to IEEE Transactions on Wireless Communications

Empowering Large Arrays from Electromagnetics and Optics

January 2023 - October 2023

Supervisor: Prof. Robert W. Heath Jr.

Motivation: To explore more efficient RoF solutions that reduce complexity and power consumption while improving scalability and to enhance system performance in digital RoF by optimizing signal processing and MIMO models for higher data rates and reliability.

Actionable Items:

- Reviewed alternative analog radio-over-fiber solutions to existing CPRI/eCPRI architecture.
- Reviewed signal processing algorithms and MIMO models for digital radio-over-fiber architectures

Distributed Joint communication and radar utilizing 5G waveform

May 2022 - July 2023

Supervisor: Prof. Robert W. Heath Jr.

Motivation: Adaptive awareness of vehicles on the road enables more efficient design and execution of applications like cruise control. In a joint multi-node scenario, where vehicles communicate directly without relying on a central unit, each node becomes fully aware of its surroundings, enhancing overall system responsiveness and safety.

Actionable Items:

- Proposed a distributed vehicular JCR feedback model for target localization.
- Developed a JCR transceiver that utilizes the 5G SSB for target detection and localization.
- "Multi-node joint communication and radar using synchronization signals in 5G" accepted at CAMSAP 2023

Multiple person Breathing Rate Estimation using RFID

Aug 2020 - May 2021

Supervisors: Dr. Momin Ayub Uppal, Dr. Muhammad Tahir

Motivation: contactless and non-invasive breath monitoring applications are becoming necessary. Given the current Pandemic, the need has increased tenfold. Utilizing the commercial off-the-shelf RFID readers, we can achieve this goal.

Actionable Items:

- Used Impinj's Low-level reader protocol toolkit to extract raw phase and RSSI values
- Created different datasets for a user in different environments, such as stationary or moving users, and multiple users
- Removed pi-jumps and noise from the phase data to extract a clean breathing signal to estimate the breathing rate

Dynamic Equivalent Circuit modelling to determine State-of-Health of Lithium-ion Battery

May 2020 - May 2021

Supervisors: Dr. Ijaz Haider Naqvi, Prof. Nauman Ahmad Zaffar *Motivation:* As Electric Vehicles are gaining popularity, the demand for efficient battery management systems(BMS) is increasing. To build a reliable BMS we need to effectively quantify the degradation due to ageing and remaining useful life.

Actionable Items:

- Model lithium-ion batteries using equivalent circuit model based on the dataset provided
- Used curve fitting tools to approximate the State-of-Health(SOH) of a lithium-ion battery.

Projects

UAV tracking using 5G-V2X joint communication and radar

Aug 2023 - Dec 2023

Motivation: 3GPP introduced 5G-V2X to improve road safety and traffic management in Intelligent Transportation Systems (ITS). In Ultra-Dense Networks (UDNs), tracking vehicular user equipment (VUE) is crucial, especially in dense urban environments where Roadside Units (RSUs) ensure constant line-of-sight, allowing accurate tracking even without GPS, enhanced by 5G's MIMO and beamforming capabilities.

Actionable Items:

- Developed simulations for localizing moving targets using an extended Kalman filter
- Designed the simulation environment with distributed MIMO stations following the METIS channel model

Maximally flat microstrip bandpass filter design using AWR

Aug 2022 - Dec 2022

Motivation: The goal of this project is to design a 10 GHz maximally flat Butterworth bandpass filter using microstrip technology with a 3 dB bandwidth of 40 MHz and 50 source/load resistance. The design uses

lossy elements, tuned to minimize insertion loss, with coupled microstrips to achieve optimal performance.

Actionable Items:

- Designed a basic lumped circuit element and then modeled it into transmission line stubs to microstrip elements.

Unsupervised abnormality detection by using intelligent and heterogeneous autonomous systems

Supervisor: Dr. Zubair Khalid *Motivation:* Automatic anomaly detection in autonomous systems is a key challenge that can open different negotiation between autonomous systems in the future. *Actionable Items:*

- Analysis and pre-processing of time series ROS Bag Dataset
- GP Regression model to predict the time series data and detect any anomaly present

Publications

S. Amir, M. R. Castellanos, Y. -H. Nam and R. W. Heath, "MIMO aided by Radio-over-multi-mode fiber for next generational RAN," preparing for submission to Transactions on wireless communications

S. Amir, M. R. Castellanos, Y. -H. Nam and R. W. Heath, "Power consumption and spectral efficiency analysis for uplink SIMO aided by radio-over-fiber," preparing for submission to Transactions on wireless communications

S. Amir, M. R. Castellanos, Y. -H. Nam and R. W. Heath, "Analog radio-over-multi-mode fiber: A spectral efficiency perspective," Submitted to ICC 2025

S. Amir, M. R. Castellanos, and R. W. Heath, "Spectral efficiency analysis of uplink MIMO aided by analog radio-over-fiber", Submitted to OFC 2025

S. Amir, M. R. Castellanos, and R. W. Heath, "Multi-node joint communication and radar using synchronization signals in 5G," in Proc. IEEE 9th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP), Herradura, Costa Rica, 2023

S. Amir, M. Gulzar, M. O. T arar, I. H. Naqvi, N. A. Zaffar and M. G. Pecht, "Dynamic Equivalent Circuit Model to Estimate State-of-Health of Lithium-Ion Batteries," in IEEE Access, vol. 10, pp. 18279-18288, 2022.

Certifications

Natural Language Processing with Classification & Vector Spaces by DeepLearning.ai

Extra Curricular & Leadership Role

- **Vice President Training and Development - Sports at LUMS**
- **Vice Captain - LUMS Netball Team**
- **General Secretary - Alpha Society LUMS**
- **Represented LUMS in 19th National Netball Championship**
- **Director Media and Promotions - LUMS Sports Fest 2020**